

Product Carbon Footprint Study Report

Product Name: washing machine

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 29, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: washing machine

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1 item washing machine

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1 item washing machine.

2. System Boundary

washing machine production in China in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of washing machine produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **466.586 kg CO₂-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

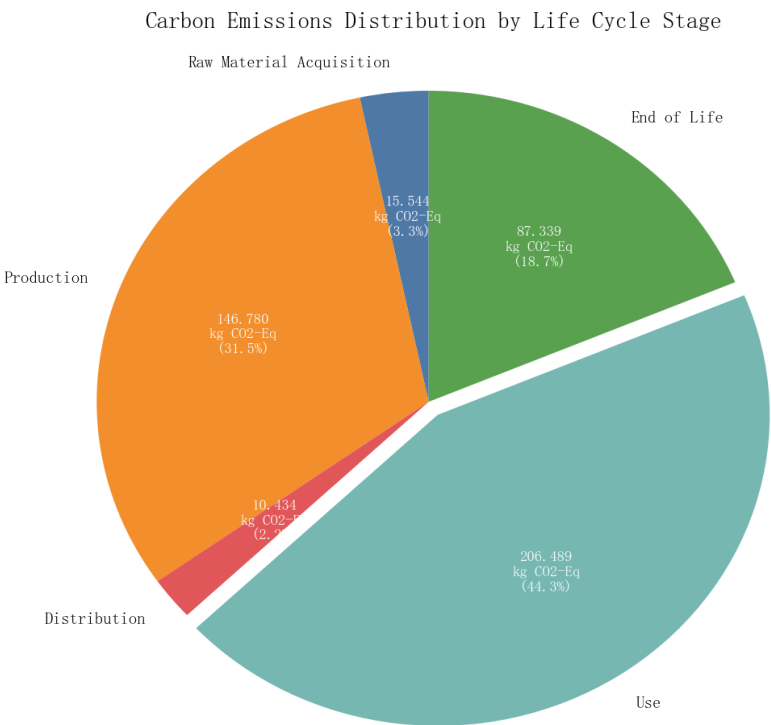


Table 1: Carbon Emissions by Life Cycle Stage

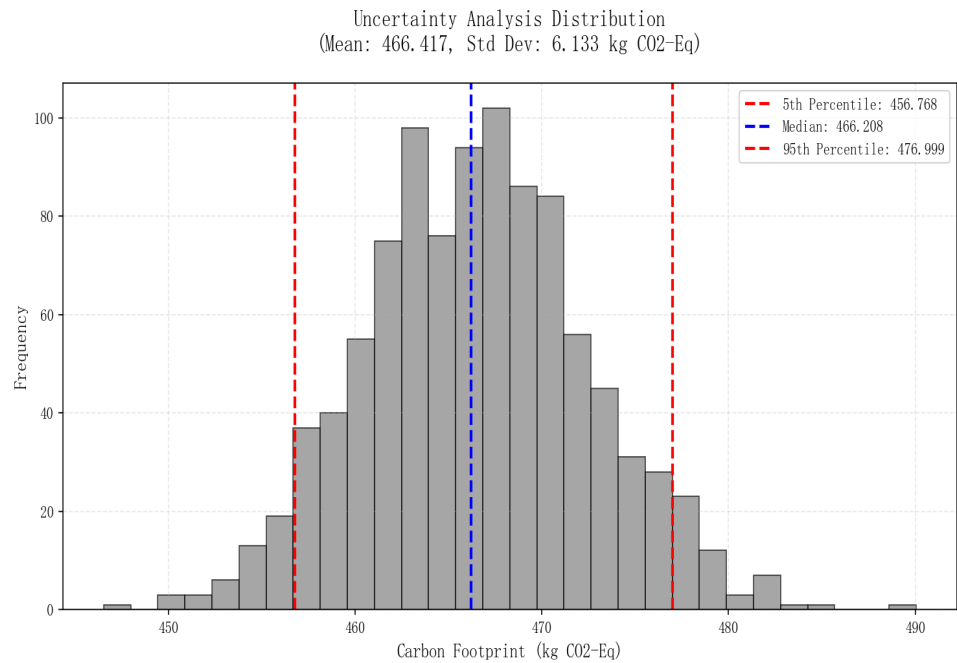
Life Cycle Stage	Carbon Footprint (kg CO ₂ -Eq)	Percentage (%)
Raw Material Acquisition	15.544	3.33
Production	146.780	31.46
Distribution	10.434	2.24

Use	206.489	44.26
End of Life	87.339	18.72
Total	466.586	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO2-Eq)
Mean	466.417
Standard Deviation	6.133
Coefficient of Variation (CV)	1.31%
Median	466.208
5th Percentile	456.768
95th Percentile	476.999

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the washing machine is 466.586 kg CO₂-Eq, which represents a moderate impact for household appliances. The lifecycle stage contributions reveal that the **use phase** is the dominant contributor at 44.3% (206.489 kg CO₂-Eq), followed by **production** at 31.5% (146.780 kg CO₂-Eq) and **end-of-life** at 18.7% (87.339 kg CO₂-Eq). Raw material extraction (3.3%) and distribution (2.2%) have relatively minor roles. This distribution indicates that the environmental performance of the washing machine is heavily influenced by electricity consumption during operation and by the manufacturing processes. The major emission sources are therefore the energy used for heating water and running the machine over its lifetime, as well as the energy and material inputs required for component fabrication and assembly. The end-of-life stage also contributes significantly, likely due to waste treatment and recycling processes. Overall, the results highlight that improving energy efficiency and optimizing production are key levers for reducing the product's carbon footprint.

2. Statement of Assumptions and Limitations

This LCA study is based on several key assumptions. First, the use-phase energy consumption was estimated assuming a standard usage pattern (e.g., 220 cycles per year, 40°C wash temperature, full load), which may not reflect all consumer behaviors. Second, the electricity grid mix was assumed to represent the average national grid, potentially masking regional variations. The data sources for raw materials and production processes are derived from industry-average databases, but the top process contributions are all listed as "Unknown" (0.000 kg CO₂-Eq), indicating a critical limitation in process-level data granularity. This absence of specific process data reduces the ability to identify and target individual emission hotspots. System boundaries are cradle-to-grave, including material extraction, manufacturing, distribution, use, and end-of-life, but exclude infrastructure capital goods and transportation of raw materials beyond the first distribution step. The uncertainty analysis (CV = 1.31%, 90% CI: 456.768–476.999 kg CO₂-Eq) suggests high precision in the total estimate, but this does not compensate for the lack of process-specific resolution.

3. Suggestions for Improvement

- **Improve use-phase efficiency:** Replace standard motors with high-efficiency brushless DC motors and promote cold-wash programs (30°C or lower). Provide clear energy labels and user guidance to reduce average wash temperature and load size.
- **Reduce production emissions:** Switch to renewable energy sources for manufacturing facilities (e.g., solar or wind) and adopt lightweight materials (e.g., recycled plastics or aluminum) to lower raw material and processing impacts.
- **Enhance end-of-life management:** Design for easier disassembly and recycling of key components (drum, motor, electronics). Partner with certified recyclers to ensure high recovery rates for metals and plastics, reducing landfill emissions.
- **Improve data quality:** Replace "Unknown" process contributions with specific supplier or industry data. Conduct a detailed process-level inventory for critical stages (production and use) to identify exact emission sources and enable targeted reductions.

- ****Encourage behavioral change:**** Include smart sensors that optimize water and energy use based on load size and soil level, and provide real-time feedback to users on energy consumption to encourage more sustainable habits.